

LaTeX Workshop Worksheet 2 – Formula

Welcome to the **LaTeX Workshop Formula Part**.

Goal: Learn the basics of LaTeX math, improve clarity and consistency, and practice writing formulas using professional conventions.

1 Common Formulas (Quick Reference)

Here are some **commonly used formulas** in mathematics, already written in LaTeX. Use them as examples and modify them in the exercises.

Remember to add `\usepackage{amsmath}` at the beginning of your code please!

1.1 Fractions & Roots

```
\[\frac{a+b}{c+d}, \quad \sqrt{x^2+y^2}, \quad \sqrt[3]{8}=2\]
```

After compilation

$$\frac{a+b}{c+d}, \quad \sqrt{x^2+y^2}, \quad \sqrt[3]{8}=2$$

Exercises

Write the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

1.2 Exponents & Logarithms

```
\[e^2 \approx 7.389, \quad e^{i\pi}+1=0, \quad \log_a b = \frac{\ln b}{\ln a}\]
```

After compilation

$$e^2 \approx 7.389, \quad e^{i\pi} + 1 = 0, \quad \log_a b = \frac{\ln b}{\ln a}$$

1.3 Limits

```
\[
```

```
\lim_{x\to0}\frac{\sin x}{x}=1, \quad
\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n=e
\]
```

After compilation

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

Exercises

Write the classic limit:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

1.4 Derivatives

```
\[
\frac{d}{dx}\sin x=\cos x, \quad
f'(x)=\lim_{h\to0}\frac{f(x+h)-f(x)}{h}
\]
```

After compilation

$$\frac{d}{dx} \sin x = \cos x, \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

1.5 Integrals

```
\[
\int_0^1 x^2 dx=\frac{1}{3}, \quad
\int_{-\infty}^{\infty} e^{-x^2} dx=\sqrt{\pi}
\]
```

After compilation

$$\int_0^1 x^2 dx = \frac{1}{3}, \quad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Exercises

Write the integral:

$$\int_0^{\infty} e^{-x} dx = 1$$

1.6 Summations & Products

```
\[
\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad
\prod_{k=1}^n k = n!
\]
```

After compilation

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \prod_{k=1}^n k = n!$$

Exercises

Write the summation:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

1.7 Vectors & Matrices

```
\[
\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad
A = \begin{Bmatrix} 1 & 0 \\ 0 & 1 \end{Bmatrix}, \quad
B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad
C = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}
\]
```

After compilation

$$\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad A = \begin{Bmatrix} 1 & 0 \\ 0 & 1 \end{Bmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad C = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

Exercises

Write the vector and the matrix:

$$\vec{v} = (x, y, z), \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.8 Set Operator

```
\[
< \quad > \quad \subset \quad \supset \quad
\subseteq \quad \supseteq \quad \cup \quad \cap
\]
```

After compilation

$< > \subset \supset \subseteq \supseteq \cup \cap$

1.9 Special Symbols

```
\[
\alpha,\ \ \beta,\ \ \gamma,\ \ \delta,\ \ \theta,\ \ \infty,\ \
\forall,\ \ \exists,\ \ \nabla
\]
```

After compilation

$\alpha, \beta, \gamma, \delta, \theta, \infty, \forall, \exists, \nabla$

Tips

1. Check [Math expressions](#) for more detailed information.
2. Use [Formula editor](#) to generate formula automatically.
3. Smart use of AI!

2 Math Modes in LaTeX

LaTeX provides **different ways** to enter math mode. It is important to use the right one depending on whether you want inline or display math.

2.1 Inline Math

- **Syntax:** `\(...\)` or single dollar `$...$`
- **Use:** Inside a sentence.
- **Example:**

This is inline math: `\( a^2 + b^2 = c^2 \)`.
Or equivalently: `$a^2+b^2=c^2$`.

After compilation

This is inline math: $a^2 + b^2 = c^2$. Or equivalently: $a^2 + b^2 = c^2$.

In professional LaTeX (academic journals), `\(...\)` is preferred because `$...$` is TeX legacy and may cause spacing issues.

2.2 Display Math

- **Syntax:** `\[...\]` or double dollars `$$...$$`
- **Use:** Standalone equation, centered, without numbering.
- **Example:**

The mass-energy equation below reveals the equivalence relationship between mass and energy: `\[ E = mc^2 \]`

After compilation

The mass-energy equation below reveals the equivalence relationship between mass and energy:

$$E = mc^2$$

2.3 Numbered Equations

- **Syntax:** `equation` environment
- **Use:** To number a formula automatically.
- **Example:**

```
\begin{equation}
\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}
\end{equation}
```

```
\begin{equation}
E = mc^2
\end{equation}
```

After compilation

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \tag{1}$$

$$E = mc^2 \tag{2}$$

2.4 Multi-line Alignment

- **Syntax:** `align` (from **amsmath**)
- **Use:** Align multi-step derivations at `=` or other operators.
- **Example:**

```
\begin{align}
f(x) &= (x+1)^2 \\
&= x^2 + 2x + 1
\end{align}
```

```
\begin{align}
f(x) &= (x+1)^2 \notag \\
&= x^2 + 2x + 1
\end{align}
```

```
\begin{equation}
\begin{aligned}
f(x) &= (x+1)^2 \\
&= x^2 + 2x + 1
\end{aligned}
\end{equation}
```

After compilation

$$f(x) = (x + 1)^2 \tag{3}$$

$$= x^2 + 2x + 1 \tag{4}$$

$$f(x) = (x + 1)^2$$

$$= x^2 + 2x + 1 \tag{5}$$

$$f(x) = (x + 1)^2$$

$$= x^2 + 2x + 1 \tag{6}$$

Exercises

Use the **align environment** to typeset

$$f(x) = (x + 1)^3 \tag{7}$$

$$= x^3 + 3x^2 + 3x + 1 \tag{8}$$

3 Best Reminder & Tips

3.1 Multi-line Alignment: align vs aligned

align

- Requires `amsmath` package: `\usepackage{amsmath}`

- Standalone multi-line equations with automatic numbering
- Example:

```
\begin{align}
f(x) &= (x+1)^2 \\
&= x^2 + 2x + 1
\end{align}
```

Notes:

- Use `&` to indicate alignment points (typically at `=`)
- Use `\nonumber` or `\notag` to remove numbering on specific lines
- Use `\label{}` for cross-references with `\eqref{}`

aligned

- Also requires `amsmath`, but must be inside another math environment
- Does not provide numbering on its own
- Example:

```
\[
\begin{aligned}
f(x) &= (x+1)^2 \\
&= x^2 + 2x + 1
\end{aligned}
\]
```

Notes:

- Only allowed in math environment. (Need `\begin{equation}` or `\[...\]`)
- Ideal for embedding aligned equations inline or inside display math
- For numbering, wrap in `equation` environment:

```
\begin{equation}
\notag
\begin{aligned}
f(x) &= (x+1)^2 \\
&= x^2 + 2x + 1
\end{aligned}
\end{equation}
```

3.2 Use `\mathrm{d}x` for Differentials

- Differential d is an operator, not a variable; it should be upright.
- Example:

$$\int_0^1 x^2 dx$$

`\[\int_0^1 x^2 \, \, \mathrm{d}x\]`

Optional macro:

$$\int_0^1 x^2 dx$$

`\newcommand{\dif}{\mathop{}\!\mathrm{d}}`
`\[\int_0^1 x^2\,\dif x\]`

3.3 Control Spacing

Command	Width	Example
<code>\,</code>	thin space (1/6 quad)	$a b$
<code>\:</code>	medium space (2/9 quad)	$a b$
<code>\;</code>	thick space (5/18 quad)	$a b$
<code>\!</code>	negative thin space (-1/6 quad)	$\boldsymbol{ab}$
<code>\quad</code>	1 em (width of “M”)	$a \quad b$
<code>\qquad</code>	2 em	$a \qquad b$

1. Multiplication

Prefer `\cdot` or thin space `\,`, over implicit juxtaposition when variables are next to numbers.

$$2x \quad \text{or} \quad 2 \cdot x$$

`\[`
`2\,x \quad \quad \text{\text{or}} \quad \quad 2\cdot x`
`\]`

2. Function Application

Always put a thin space after functions before arguments for clarity.

$$\sin x, \quad \log n$$


```
\[
\sin\,x,\quad \log\,n
\]
```

3. Fractions

Use small spaces around `\frac` if needed for readability.

$$\frac{a}{b}x$$

```
\[
\frac{a}{b}\,x
\]
```

4. Operators in Limits/Integrals

Add thin or medium space around d in integrals.

$$\int_0^1 f(x) \, dx$$

```
\[
\int_0^1 f(x)\,,\mathrm{d}x
\]
```

5. Negative Space

Use `!` sparingly to reduce awkward gaps in products or superscripts.

$$x^{n+1}$$

```
\[
x^{n\!+1}
\]
```

To see more detailed information in section 7 of this webpage: [latex.space](#).

3.4 Place Limits Explicitly

- Inline math usually puts limits to the right: $\sum_{i=1}^n i$
- Use `\limits` to force display-style limits inline: $\sum_{i=1}^n i$
- Or `\displaystyle`: $\sum_{i=1}^n i$

```
\(\sum\limits_{i=1}^n i\)
```

```
\(\displaystyle\sum_{i=1}^n i\)
```

3.5 Use `\label` and `\ref` for Numbering

- Automatic numbering and cross-references
- Example:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{9}$$

As shown in equation (9), ...

```
\begin{equation}
\label{eq:problem1_quadratic}
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{equation}
```

As shown in equation `\eqref{eq:problem1_quadratic}`, ...

- `\ref{eq:problem1_quadratic}` → prints just the number: 1.
- `\eqref{eq:problem1_quadratic}` → prints the number with parentheses: (1).

4 Other Related Materials

1. [AutoTableGenerate](#)
2. [The Comprehensive TeX Archive Network](#)